Power BI Dashboard: Global Population Analysis

A comprehensive analytics solution built with Power BI, Python, and Wikipedia data to track and analyze global population trends across countries and time.

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Project Overview & Objectives

What We Built

A dynamic Power BI dashboard that visualizes population data from Wikipedia, enabling users to explore demographic patterns, compare countries, and track historical trends with interactive filters.

Core Objectives

- Centralize global population data in one accessible view
- Enable quick identification of population leaders and patterns
- Support data-driven decision making through clear KPIs
- Provide historical context for population changes

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Data Collection Process

01

Data Source

Wikipedia population pages, chosen for comprehensive coverage across all countries and consistent historical data availability.

02

Web Scraping

Python scripts using BeautifulSoup and Requests libraries to extract tables, country names, population figures, and year data automatically.

03

Data Export

Raw data saved to CSV files for import into Power BI, maintaining source integrity while enabling transformation.

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Data Processing & Transformation

Power Query Operations

Data Cleaning

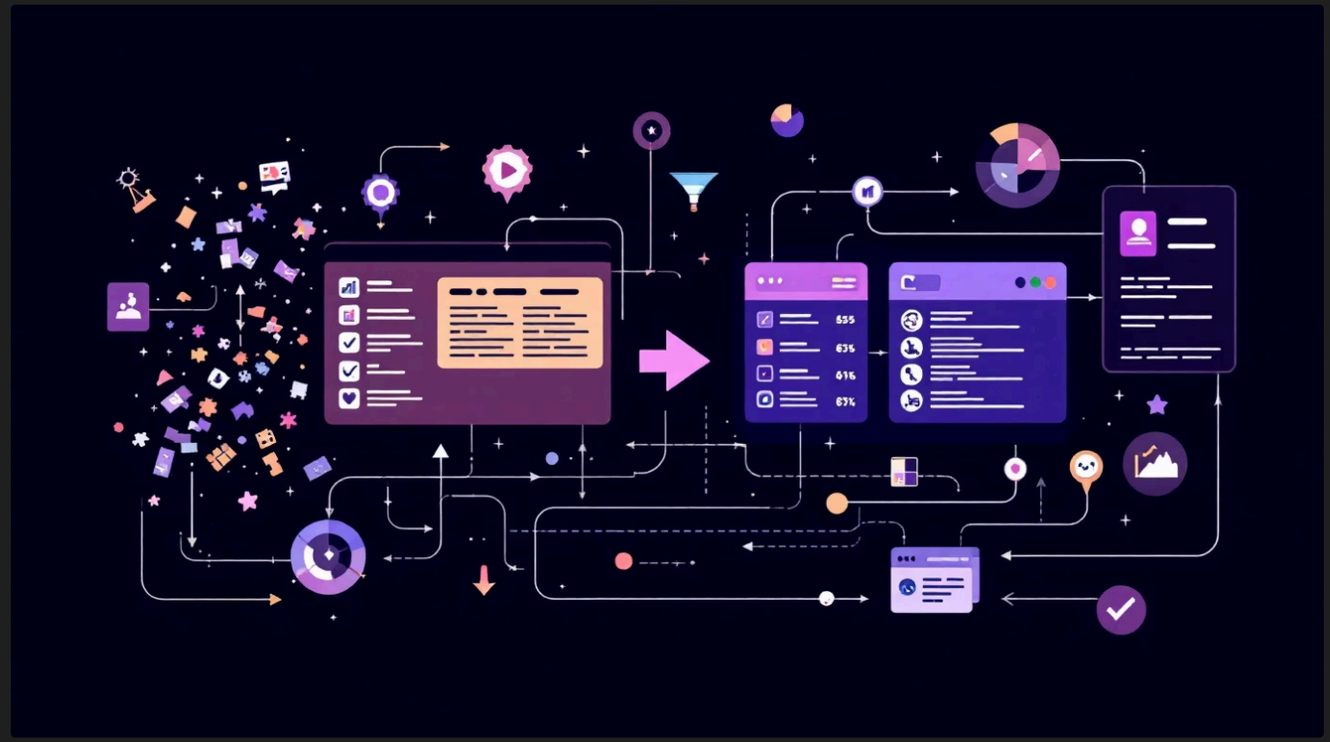
Removed duplicates, standardized country names, and validated number formats across all records.

Missing Values

Applied forward-fill logic for historical gaps and flagged incomplete country data for manual verification.

Format Conversion

Transformed percentage strings into decimal values, ensuring consistent numeric calculations throughout.



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Key DAX Measures

Total Population

Sum of all country populations for a given time period

Average Population

Mean population across all countries in the dataset

Maximum Population

Identifies the most populated country at any point in time

Minimum Population

Finds the least populated country for comparison

Population %

Calculates each country's share of global population

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Dashboard Overview Page

Key Performance Indicators

Top-left positioned KPIs display Total, Average, Maximum, and Minimum population values for immediate visibility.

Top 10 Countries Bar Chart

Horizontal bar chart ranks countries by population, highlighting the largest nations at a glance.

Population Distribution

Pie or donut chart shows percentage contribution of each country to total global population.

Interactive Date Filter

Slicer control enables users to explore population snapshots across different years.

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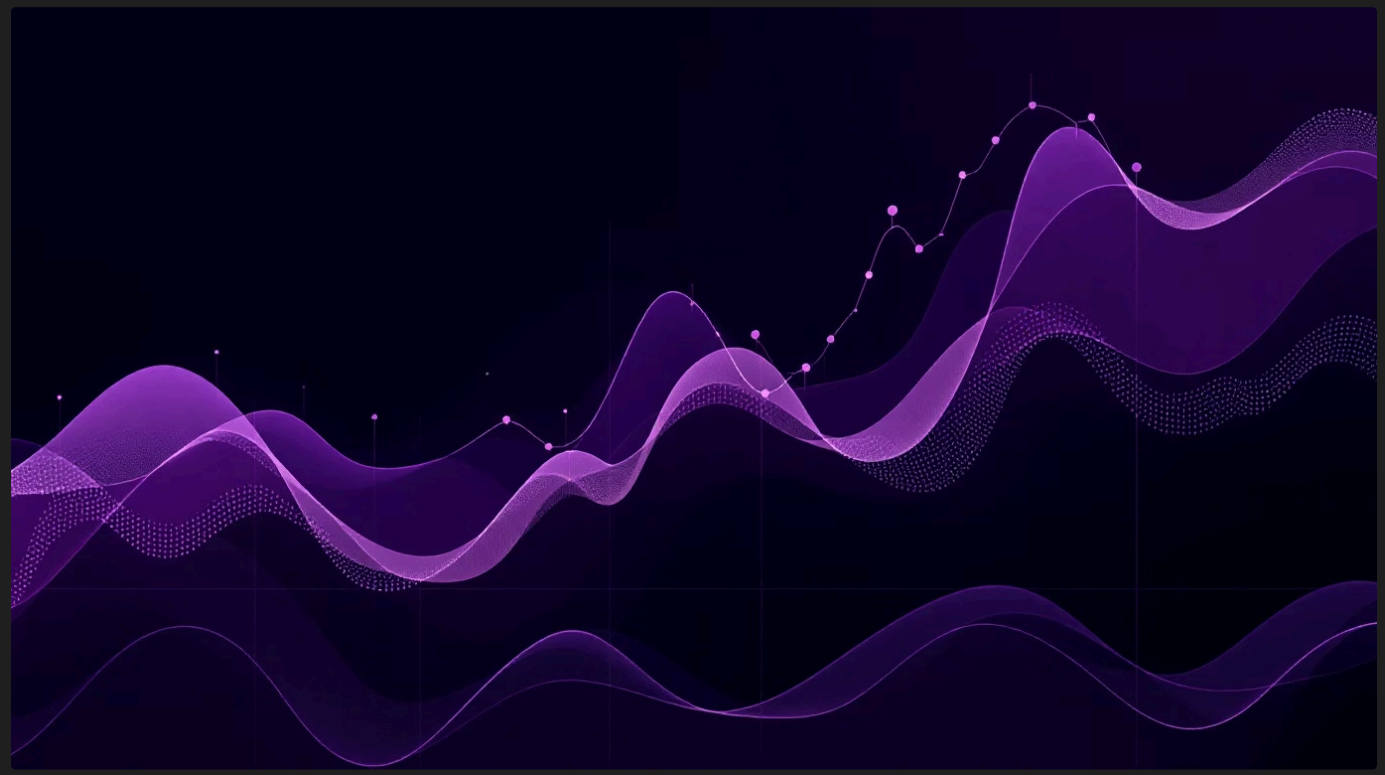
Analysis Page Features



Population Trend Visualization

Line chart plots population changes over time, revealing growth patterns, inflection points, and long-term trajectories for selected countries.

Comparative Analysis

Side-by-side comparison of Top vs Least populated countries highlights the massive disparity in population distribution across the globe.

Interactive Filtering

Dynamic slicers and cross-filtering allow users to drill into specific regions, time periods, or population ranges instantly.

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Key Insights Revealed

Population Leaders

China and India dominate the top positions, collectively representing over one-third of global population. Their growth trajectories differ significantly over time.

Distribution Patterns

Asia accounts for 60%+ of world population, while Africa shows fastest growth rates. Europe and Americas remain relatively stable.

Historical Trends

Global population crossed 1 billion in 1800, 7 billion in 2011, and now exceeds 8 billion. Growth accelerated dramatically in the 20th century.

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Tools & Technologies

Power BI

Primary platform for data modeling, DAX calculations, and interactive visualizations

Python

Backend scripting for web scraping and data extraction automation

Pandas

Data manipulation library for preprocessing and transformation tasks

BeautifulSoup

HTML parsing library that extracts structured data from Wikipedia pages

Requests

HTTP library for sending web requests and retrieving page content

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Conclusion & Value Delivered

Project Success

This dashboard successfully transformed Wikipedia's raw population data into an interactive, business-ready analytics tool that answers real questions about global demographics.

Key Achievements

- Automated data collection from Wikipedia
- Clean, reliable dataset with consistent formatting
- Dynamic visualizations that update with new data
- Intuitive interface requiring no training

Business Value

Stakeholders can now make informed decisions about market sizing, resource allocation, and strategic planning based on accurate population intelligence.

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